

than several weeks. Given these problems, it is natural that corporations are progressively employing more technological innovations.

Advances over the decades

Virtualisation, a technological innovation long associated with mainframe computer systems, has been changing IT facilities due to its capability to consolidate hardware resources and decrease energy costs. This has led it to grow as a practical technological innovation for mobile phones and exclusive personal systems, as well as being used to reconceive nimble and cloud computing.

Virtualisation, in this new, effective era of cloud computing, is pushed by the need for spending budgets effectively, for agility and for meeting other challenges in the traditional environment.

In the 1960s, the time-sharing of systems was preferred to batch processing systems. Virtualisation became a means to fully utilise hardware components and assist in the optimum use of systems on a time-sharing basis.

The phrase 'hypervisor' was first used in 1965, mentioning software that associated an IBM RPQ for the IBM 360/65. A hypervisor or Virtual Machine Monitor enables multiple guest operating systems on a host computer.

TYPE 1 / NATIVE / BARE METAL HYPERVISORS	TYPE 2 / HOSTED HYPERVISORS
As per Wikipedia, these run directly on the physical host to control the hardware and to manage guest operating systems. They represent the classic implementation of virtual machine architectures.	As per Wikipedia, these run within a traditional operating system environment. With the hypervisor layer as a distinct second software level, guest operating systems run at the third level above the hardware.
Examples: Citrix XenServer, VMware ESX/ESXi, and Microsoft Hyper-V hypervisor.	Examples: KVM and Virtual Box

In the mid-60s, IBM's Cambridge Scientific Center developed the CP-40, the first edition of the CP/CMS. It went into production in January 1967 and was designed to implement full virtualisation. IBM mainframes have supported several completely virtualised operating systems since the early 70s.

In the 80s and into the 90s, virtualisation was mostly overlooked, as affordable PCs and Intel-based hosts became popular. Over time, the expenditure on physical facilities, failover and catastrophe protection needs, the high cost of systems servicing, and low server utilisation became problems that required a new solution.

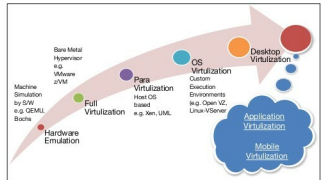


Figure 1: Advances in virtualisation over the decades

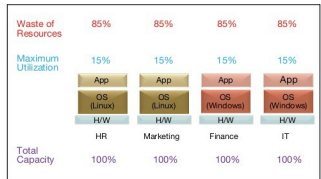


Figure 2: Resource utilisation before virtualisation

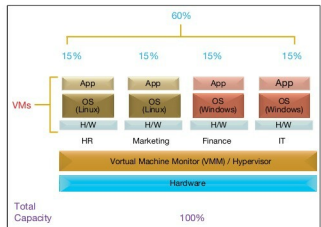


Figure 3: Resource utilisation after virtualisation

In the late 90s, x86 virtualisation was achieved through complicated software techniques, which overrode the processor's lack of virtualisation support and accomplished affordable efficiency. Virtualisation of Intel-based devices became a potential solution. This was followed by the arrival of VMware, which overcame the hardware limitations that blocked virtualisation on Intel-based structures. Since then, virtualisation achievements have led to what could be called a virtualisation rebirth these days. The current generation is discovering what had already been done long ago, but is implementing the advantages in the present technical landscape. In 1999,